

1 Status Assignment

HYDRA menu	Master data → Workplaces/ machines → Status assignment
FEDRA menu	Detailed scheduling → Master data → Status assignment
Transaction code	mst
Function authorization	mdmst

You can create statuses for all workplaces/ machines. The status shows the current status of the workplace/ machine.

Purpose

All possible statuses or malfunctions at the machine/workplace are configured in the *Status assignment* and assigned to the status texts by distinct status numbers. If malfunctions are detected on the terminal, the system uses the statuses documented in the status assignment.

Example of how a status table is set up :

Status	1	2	3	4	5	6	12
Status text number	1	3	4	5	6	2	12
RPA	MUT	LCI	SCI	SET	BKS	DCI	DCI
Control indicator	Production	Malfunction	Malfunction	Malfunction	Malfunction	Malfunction	Gen. disturbance
Manual assignment via the terminal (Manually at terminal)	☐	☐	☐	☒	☒	☐	☒
Automatic assignment via digital input	☒	☒	☒	☒	☐	☒	☐
Digital input	1	2	3	4	0	6	0
Assignment on the machine interface	0	2	1	3	-	7	-

The entries are based on the status texts listed below:

- 1 Production
- 2 Tool breakage
- 3 No raw material
- 4 Staff shortage
- 5 Setup
- 6 Break
- 12 Gen. disturbance

This is only an example illustrating the status assignment.

The selection dialog allows you to select and view the statuses that have already been assigned to a machine or a workplace.

Integration

The workplace/ machine status is integrated in various evaluations/reports.

Requirements

You have to create the following objects before you can assign workplace/ machine statuses.

- Workplace/machine
- Status text
- Status class (optional)
- Resource Performance Account (RPA)

Toolbar



Status list

Click the "Status list" icon to open the report including the defined and selected statuses.

This report shows the statuses in printable form as plain text and as bar code. Click here

[Machine status list report](#) to find detailed information.

Selection criteria

The application provides the following selection criteria:

Workplace

Select the workplace/machine for which you want to display the statuses.

Status

Select the defined workplace/machine statuses. The application shows the statuses of all workplaces/machines matching the entered status number.

Status text

Select the defined status text. You can also use wildcards.

RPA abbrev.

Abbreviation for the assigned Resource Performance Account

Status class

Abbreviation for the assigned status class

Field descriptions



Notes on status 30000

You can only change selected options for status 30000:

- Warning in the graphic machinery
- Activate production lock
- Activate machine lock
- Record scrap reasons

Certain fields such as status class, RPA, control indicator are not filled with this status and are therefore empty in the list.

General tab

Machine/workplace (short name)

Enter the machine number to assign the disturbance status to a machine. Enter the same number in this field for all statuses of a machine in order to create a complete status table for a machine.

The *short name* includes the name of the entered or selected machine.

Status

This field includes the unique number for statuses included in the status table. You can also use this number to assign and/or change the status via the terminal.



You can only define one status for workplaces of the type "group workplace". Assign the characteristic "production" (see "control" tab) to this status.

Note: You cannot delete the status 30000 "Not assigned". You can only configure specific fields for this status.

Superior status

If no status is defined in the Superior status field, the currently created status is at the highest level.

Otherwise, enter the number of the directly superior status. This must already exist and have the control indicator *Hierarchy level*.

Note: This function is available only for Windows terminals.

Status text

The number assigned here refers to the plain text status message from the status text table.

Status class

Assign a status to a status class to make cumulative evaluations/reports about status classes. The abbreviation assigned here refers to the plain text status class message from the status class table.

Resource Performance Account (RPA)

Enter a value in this field to assign the status to a Resource Performance Account (RPA). Select one of the 12 Resource Performance Accounts (RPA).

By default, the 12 Resource Performance Accounts are already defined in HYDRA. Refer to the glossary for further information on the Resource Performance Accounts.

Control tab**Control indicator**

The following characteristics are available to specify machine monitoring.

Except for the "production" characteristic, all following characteristics are only allowed for machines/workplaces of the type "individual workplace".

Production

Production identifies the production state/status of a machine/workplace. Assign the "production" characteristic to **exactly one** status for each workplace.

If the machine monitoring system detects production signals, then the status of machines/workplaces of the type "single workplace" is changed to the status to which this control indicator is assigned.

Only one status is allowed for workplaces of the type "group workplace" and this status must be assigned this control indicator.

Other status

Assign the control indicator "other status" for all statuses without a control indicator. You can assign this control indicator/characteristic to any number of statuses for each individual workplace.

General disturbance

Create exactly one status as general disturbance for each machine/workplace. If the machine data collection detects a production phase that has not yet been assigned a disturbance or status, then this duration is posted to the status assigned to the control indicator for the general disturbance.

Material change

This option is only available with the MPL module.

If a workplace has a status with this indicator, materials that are not planned can also be logged on. These materials are logged on as alternative material, which means that you log on some other material instead of the planned material. You can specify in the status assignment if alternative materials may be logged on. As a result, you have to configure at least one "Setup" status with the "Material change" option in order to enable the desired posting behavior. If the machine is in a status that is assigned the "material change" option, you can log on the alternative material already when you log on the operation. In this case, you do not even have to change the status beforehand.

No order

If you interrupt or log off an operation manually, the Windows terminal verifies whether this is the last operation of the workplace. If no more operations are logged on to this workplace, the terminal sets the workplace status to the status assigned to the "No order" option.



This option is only available for Windows terminals (CTWIN/AIP).

A status with this control indicator may only be configured at machine/workplace of the type "single workplace". You may configure only one status with this control indicator.

Do not configure a status with the control indicator "No order" at workplaces of the type "group workplace", since only the status "Production" is available at group workplaces.

Short-term status (as of MDE 7.2)

For an optimized overview, for example in the status log or machine history, you can configure one status per machine as a short-term disturbance. Use this status as a "repository" for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal identifies a downtime and the machine automatically returns to the status "production", the system verifies if this disturbance took less time than configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

The system does not differentiate between such automatic status postings (automatic assignment of reasons) for short-term disturbances and reasons entered manually by operators. The duration of short-term disturbances is ignored with automatic shift changes.

Hierarchy level

Assign the option "hierarchy level" to statuses that cannot be recorded via the terminal. These statuses only represent the hierarchy.



This function is only available for Windows terminals. In this case, you can neither change specific configurations nor enter data in certain tabs.

Estimated downtime

If you assign a status manually, you can enter an estimated downtime. In this case, the application suggests the downtime that is stored in the master data.

If statuses are changed automatically, the system assigns the downtime stored in the master data automatically.

Activate production lock

If this option is set, the production lock (P lock) is automatically activated when a status is assigned via the terminal.

If you use the machine monitoring function, the *production lock* option prevents the machine from switching automatically to the status "production" when a production signal arrives. Consequently, this status overrides the production signal until you manually disable the production lock option. The production lock can also be used to determine whether and how quantities (counter readings) are posted during this time.

Setting the machine lock output

You have to set this option, if you want the machine lock to be enabled when assigning statuses.

In this case, you also have to ensure that the machine lock output has been configured accordingly in the machine configuration.

Warning in the graphic machinery

The entry determines the time after which the symbol (more precisely: the part of a symbol that represents the status in color) starts flashing in the [Graphic machinery](#) after the workplace/machine status has occurred. Enter the value in the format hours: minutes: seconds.

Status change tab**Manually via the terminal**

If this option is selected, you can enter the status manually via the terminal (using barcode or keyboard). If this option is not selected, the status selection list of the terminal will no longer show the status.

Authorization

Access authorization for entering a status via the terminal (enter a value between 0 and 9). An authorization level for machine status modification is defined for every person in the HR master data. If the authorization level stored in the master data is lower than the authorization level defined here, you cannot assign the status via the terminal.

Automatically via digital input

Select this option, if you want the statuses to be assigned automatically via the machine interface (CT-MSS, CT-UMPS, PCC). Enter the number of the digital input identifying the status in the "digital input" field (0 = no input).

If you monitor machines via the operating signal, a digital input also records the operating signal. Proceed as follows to define a status as an operating signal:

- Control for machine monitoring: "Production"
- Select the option "Automatically via digital input"
- Enter a value > 0 in the "digital input" field. This input records the operating signal.

In the case of disturbance reasons, a general distinction is made between automatically and manually recorded disturbance reasons. Distinguishing features are:

- Disturbance reasons you enter manually at the terminal override automatically set disturbance reasons.
- Operating signals do not override a status you set manually. Except for operating signals for the status "production" (see next bullet point). The "production" status also overrides a manually set status.



If no production lock is set, the status with the control indicator "Production" overrides every disturbance reason. Therefore, keep in mind that the status "Production" must be deactivated, if you want a current disturbance to be processed at the input.

If multiple automatic statuses (disturbance reasons) are recorded via digital inputs, the status with the lowest HYDRA channel number (not the lowest status number!) is set.



Note that the assignment of the number to the physical connection at the MSS depends on the settings in the local terminal configuration file (Windows terminal: CTWIN.INI/CTAIP.INI, DOS terminal: AIOP.CFG).



If a status is to be processed automatically by the machine based on the transferred HYDRA status number (MSTAT), this option "Automatically via digital input" must be activated and the field "Digital input" must be assigned the value "0".



If a digital input and a status number (MSTAT) for a machine status are set at the same time for a machine, the status of the digital input at the machine is set. The digital input therefore has a

higher priority than the status via MSTAT.

Digital input

Number of the digital input used to set the status.

Status transfer to aggregates

This option allows you to set a global status at the production line, which is then automatically set for all aggregates assigned to it.

The requirement for this is that this status is also configured for all assigned aggregates and that the status number for the aggregates is identical to the status number of the production line.

Processing tab

Log off staff

Please note: the weekend automatic function (status 999) does not support this option.

If you select this option, the system logs off all persons currently logged on when a status is assigned (useful during maintenance phases). Otherwise, the persons stay logged on.

Operation posting

Please note: the weekend automatic function (status 999) does not support this option.

Use the option "operation posting" to have an overhead cost order logged on automatically when statuses change. The following options are supported:

None

No processing.

Interrupt OP

Use this setting to interrupt automatically all active operations and to log off all employees from the workplace if this machine status is set.

Interrupt active OP and log on the following OP

Use this option to interrupt all active operations and to log off all employees when statuses are changed. The system automatically logs on the "subsequent operation" stored in the *Operation* field.

Please note: The subsequent operation must not be subject to batch management.

Transfer registered persons to OP

Depending on this setting, the system transfers the persons logged in to the "subsequent operation" defined in the *Operation* field.

If the operation is an overhead cost operation or if the workplace is a group workplace (GAP), then at least one person is logged on to the subsequent OP:

- either the person carrying out the posting, or
- the person who is logged on the longest if the *change status* dialog does not include a field to enter the staff badge number. If no employee is logged on at this time, the subsequent operation cannot be logged on.

Scrap reason

Depending on the current status, you can post automatically recorded scrap to a defined reason.

A distinction is made in the process, whether the production lock is set or not while the status is active. You can choose from the following configuration options:

- scrap reason
- scrap reason during production lock

A counting input explicitly defined as a scrap counter generally takes priority over a reason defined here.

Plausibilities tab

Check running operation

You can configure statuses,

- | | |
|--|--|
| - if an OP is logged on to the workplace | Use Option <i>An operation must be logged on</i> |
| - if no OP is logged on to the workplace | Use Option <i>An operation must not be logged on</i> |
| - if an operation is logged on to the workplace or not | Use Option <i>No check</i> |

HYDRA checks this dependency during the manual status change and during the manual login/logout or interruption of operations. If the posting violates the condition, the terminal issues an error message and refuses the posting.



If a new status is set when an operation is terminated or interrupted, then this status cannot be an order-related status!

Use of unplanned material allowed

This option is relevant if you use the HYDRA module Material and Production Logistics (MPL). If the option is set, you can use unplanned material. That means, you can log on batches that are not specified in the input material list of the operation as input material. This can be useful during setup.

Tab *User fields*

User field key

This field of the object type MST is preset with the user field key DEFAULT. Normally, you cannot change this assignment. MPDV defines the user fields for this user field key during the customizing process to meet specific customer requirements.